

Changes in Preferential Flow Path Distribution and Its Affecting Factors in Southwest China

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Abstract: Preferential flow is important in solute transport in the soil, and it indirectly affects the quality of the groundwater. The objectives of the present study were to investigate the distribution characteristics of preferential flow paths under different land-use types (forestland, shrubland, and farmland) and to determine the factors that affect the formation of these paths. Dye tracer experiments were conducted at six sites covering the three land-use types. The relationship of the preferential flow paths to the properties of the soil, root length per soil volume, root holes, and root surface area was analyzed using a partial correlation method. The results show dye coverage of greater than 50% at a soil depth of 0 to 10 cm and dye coverage of less than 30% below a depth of 20 cm in all of the land-use types. The vertical distribution differed widely in the various land-use types. Shrubland and farmland had more preferential flow paths in the topsoil compared with forestland, in contrast to the conditions in the deeper soil layers. The soil chemical properties and the saturated hydraulic conductivity in the preferential flow paths were quite different from those in the soil matrix. The soil organic matter content was approximately 5% to 20% higher in the preferential flow paths than in the soil matrix. The average value of the saturated hydraulic conductivity was 41.3 mm min^{-1} in the preferential flow paths compared with 21.4 mm min^{-1} in the soil matrix. The percentage of preferential flow paths in the soil was significantly affected by roots with a diameter of less than 5 mm, root holes, and the root surface area. These findings suggest that land use is significant in the distribution of preferential flow paths, and root characteristics are important in the formation of preferential flow paths.

Key words: Preferential flow path, image analysis, soil properties, root characteristics, land use.

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Preferential flow refers to the uneven and often rapid movement of water and solutes through a porous medium, typically soil. This movement is characterized by regions of enhanced flux such that small media (such as wormholes, root holes, and cracks) participate in most of the flow. This mechanism has received considerable attention because it accounts for a significant increased risk of groundwater contamination from the soil (Germann and Dipietro, 1996; Stamm et al., 1998; Lipsius and Monney, 2006; Ronkanen and Klove, 2009).

Preferential flow has been observed in all but those soils with the coarsest texture, which conforms to the uniform flow theory (Mooney and Morris, 2008). In saturated soil, preferential flow involves rapid water displacement from macropores (hydrodynamic dispersion), whereas in unsaturated soil, flow into air-filled macropores (short-circuiting) occurs, followed by a lateral absorption (Weiler and Naef, 2003). Pesticide transport along preferential flow pathways was a major contributor to the total pesticide displacement (Reichenberger et al., 2002). Preferential flow paths prevailed in most soil profiles, with the dye coverage ranging between 8% and 40% (Lipsius and Monney, 2006).

The causes and extent of preferential flow have been studied (Flühler et al., 1996; Weiler and Naef, 2003). Glass and Nicholl (1996) addressed three complex factors that fundamentally alter the gravity fingering process, namely, the uniform and nonuniform initial moisture content, the media heterogeneity, and the presence of macropores and fractures. Preferential flow paths have not been found in areas with high water repellency and very low water content; thus, preferential flow may be prevented by water repellency (Weiler and Flühler, 2004). Soil texture is also a major factor in the formation of preferential flow paths. The experimental results of Novak et al. (2001) indicate that metolachlor losses in drain water are primarily caused by preferential flow, which is greater in clay soil than in silt loam, and it occurs mainly in the spring and summer.

Root decay has been identified as the cause of macropores involved in the preferential flow under cropping conditions, and root systems increased infiltration rate (Barley, 1954). The presence of roots also improved preferential infiltration and helped develop preferential flow paths (Lange et al., 2009).

Earthworm activity causes changes at different scales of soil porosity, including macroporosity (Lamande et al., 2003), and preferential herbicide transport through earthworm burrows has been observed during rainfall simulations (Farenhorst et al., 2000). However, earthworm activity cannot produce stable macropores (Mitchell et al., 1995). Both freeze-thaw and wet-dry cycles can cause macropores and preferential flow in soils, but the former is stronger than the latter. Land use also greatly affects the formation of preferential flow paths because root distribution and soil properties are quite different under various land-use types. Malone et al. (2003) showed that differences in the sub-surface soil properties (i.e., 8–30 cm) can cause percolation through macropores earlier in no-till (NT) croplands than in moldboard plowed croplands and higher herbicide concentrations in the percolates of NT croplands.

Dye tracer experiments are usually used for the visualization, identification, and quantification of water flow pathways in structured soils (Flury and Flühler, 1994; Ohrström et al., 2004). Brilliant blue FCF has been widely used for this purpose because it is nontoxic, is distinctly visible in soils, and can be considered moderately mobile at large concentrations (Flury and Flühler, 1995; Kasteel et al., 2002; Mon et al., 2006). Abou Najm et al. (2010) used liquid latex to quantify the soil preferential paths and to assess the evolution of the preferential

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paths in the field, a method that has not been widely used until recently.

Preferential flow is a common phenomenon in Southwest China, where it is affected by the soil properties (Cheng et al., 2007). The distribution variety of the preferential flow paths in different forestlands has been noted (Wang et al., 2010), but the land use and root characteristics have seldom been recorded in relationship to these flow paths.

The objectives of this report were the following: (i) to describe the distribution of the preferential flow paths under different land-use conditions and (ii) to evaluate the effects of land use and parameters, such as the soil properties, root length, root holes, and root surface area, on the formation of the preferential flow paths.

MATERIALS AND METHODS

Study Site

As a fragile and important ecological transition zone, the Three Gorges Reservoir Area, situated in the upper reaches of the Yangtze River, represents the complexity of the natural environment in China. Simian Mountain is at the upper end of this area. The field site, the Qinjiagou watershed, is located in the middle part of Simian Mountain in Southwest China (28°31'–28°46' N, 106°17'–106°30' E) (Fig. 1). The elevation of the research sites ranges from 900 m to 1,500 m above sea level. Simian Mountain is located in a sub-tropical area and has a continental monsoon climate with abundant precipitation. The annual average precipitation is 1522.3 mm, 98% of which falls as rain, occurring mainly from May to September. The annual average runoff is 675 mm, with 56.1% occurring from April to July. The average annual temperature is 13.7°C, with

the lowest monthly average temperature (−5.5°C) in January and the highest (31.5°C) in August. The average soil depth is about 50 cm.

To address the effects of land use on the preferential flow paths, two forest sites (hardwood and softwood), two farmland sites (tilled and nontilled), and two shrubland sites were investigated in the Qinjiagou watershed. The locations of the six sites are shown in Fig. 1. The main tree species at the hardwood forest (HF) site were 10-year-old sweetgum (*Liquidambar formosana*) and camphor (*Cinnamomum camphora*). The main tree species at the softwood forest site (SF) was Chinese fir (*Cunninghamia lanceolata*). In the experiments, most of the fallen leaves were removed, leaving behind an undisturbed litter layer. The crop at the tilled (T) site was corn; the experiment was conducted just after harvest, and the soil surface was bare with little plant residue on top. The NT farmland had not been tilled for 2 years at the time of the experiments. Soybeans had been planted in the farmland before it was left untilled. Both shrub sites (S1 and S2) were located in a 10-year-old shrubland. The main shrub types were *Rubus amphidasys*, *Rubus corchorifolius*, *Rubus hunanensis*, and *Rubus rosaefolius*. The most part (90%) of the shrubland surface was covered by grass, the coverage of grass in the farmland is about 10%, and the forestland is about 40%. The plots were slightly sloped, and to minimize the influence of variables other than land use, we chose the sites that displayed similar soil textures (Fig. 2). The initial soil-moisture content was similar in all of the sites.

Macropore Flow Path Detection

Image analysis with dye tracers is the most widely used method to describe the distribution of preferential flow paths. As a dye tracer, brilliant blue FCF is widely used to visualize flow

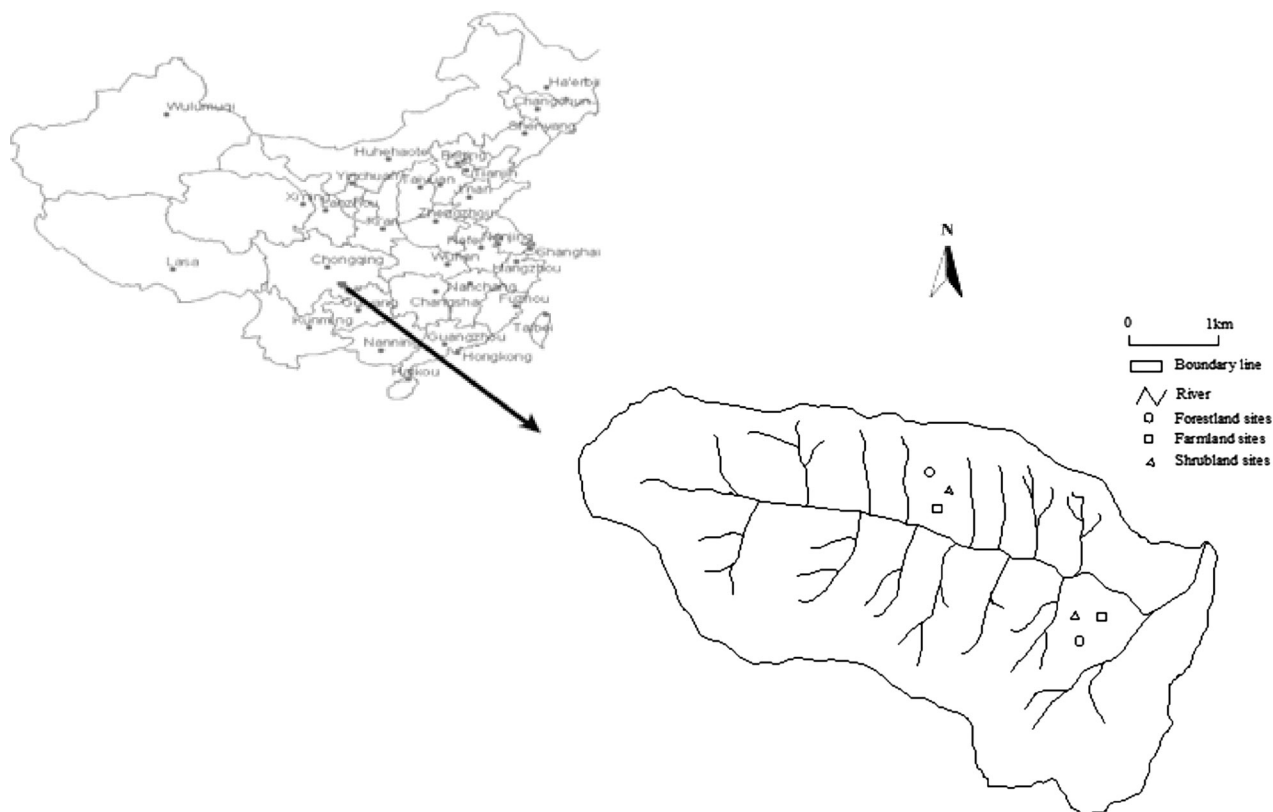


FIG. 1. Location of the experimental sites.

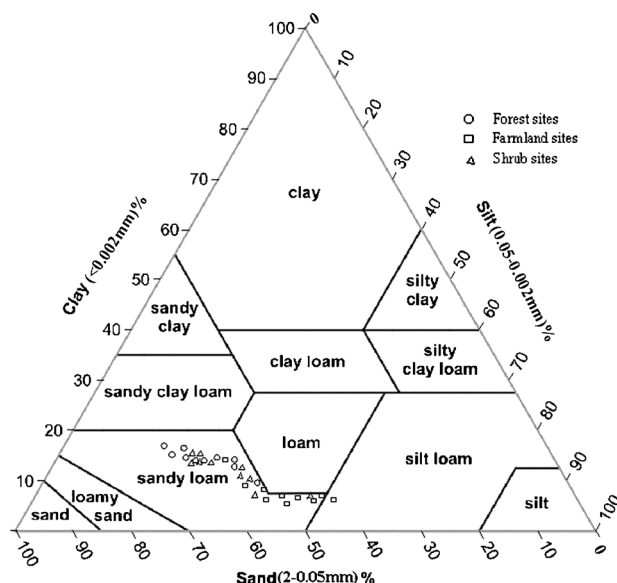


FIG. 2. Soil texture in the different land-use types.

paths (Flury and Flühler, 1995); accordingly, this dye was used to stain the preferential flow paths in the present study.

The tracer experiments were conducted in October 2008. At each site, a 100×100 -cm flat area was selected for the dye experiment; an iron frame with a volume of $70 \times 70 \times 50$ cm (length $\times$ width $\times$ height) and thickness of 0.5 cm was embedded into the soil at a depth of 30 cm. The soil surface within 5 cm at the inner and outer sides of the iron wall was compacted to keep the dye tracers from leaving the frame (Fig. 3). Water (10 L) was dyed with brilliant blue FCF at a concentration of 4 g L^{-1} , and the tracer application was conducted manually with a backpack sprayer of the type that is normally used for pesticide application. The application rate was determined by the infiltration capacity of the soil because ponding was expected to be prevented, yet it still occurred. The application took 1.2 h for the forest plots, 1.0 h for the shrub plots, and 1.1 h for the farmland plots; after 24 h, a 0.7-m-deep trench was dug beside the dye application area. Then, the experimental plot was dug up in 5 horizontal cross sections, each with a depth of 10 cm (Fig. 3). A border of 10 cm along the frame was not taken into account for image analysis, namely, the area of horizontal cross section was 50×50 cm (length $\times$ width). In the present study,

the dye-stained area was defined as the preferential flow path, whereas the dye-unstained area was considered the soil matrix.

The horizontal cross section of each layer was captured by a digital camera ($2,560 \times 1,920$ pixels). Using Adobe Photoshop CS3 (Adobe Systems Incorporated, San Jose, California), the perspective effect of these digital images between the camera lens and the soil layer surface was corrected. The dye-stained areas were black and gray, whereas the unstained areas were white. For each of the horizontal cross sections, the average dye coverage was calculated and converted to a gray-scale value using Image-Pro Plus 6.0 (Media Cybernetics Incorporated, Silver Spring, Maryland).

The characteristics of the preferential flow paths at each site, including the color, shape, and size, were also assessed. Some of the preferential flow paths were quite flat at their inner side, and some had residues of decayed and semidecayed roots. The preferential flow paths with the flat inner sides were possibly caused by animal activity (Lamande et al., 2003). In the field, no other animals were found in the soil except live earthworms; therefore, this kind of macropore was inferred to be formed by earthworm activity. The macropores with root residues were defined as root holes. However, there were few earthworms at the shrubland sites and even fewer at the forestland and farmland sites. At the forestland sites, most of the macropores were root holes. At the farmland sites, some 5-mm-wide cracks and fissures were also found in the topsoil, but few root holes were observed.

In the current study, the relationship between the preferential flow paths and the general factors caused by different land-use types was analyzed.

Soil Sampling and Analysis

A total of 2 soil samples ($10 \times 10 \text{ cm}^2$) in each soil layer of the six experimental sites were taken from the dye-stained and dye-unstained areas for nutrient measurements. The chemical properties of the soil samples were also analyzed. The soil organic matter was determined using an oil bath- $\text{K}_2\text{Cr}_2\text{O}_7$ titration method. The available phosphorus was extracted with a $0.5 \text{ mol L}^{-1} \text{ NaHCO}_3$ solution (pH 8.5). The rapidly available potassium and total nitrogen were determined using a $\text{CH}_3\text{COONH}_4$ extraction method and the semimicro Kjeldahl method, respectively (Editorial Committee, 1996).

Two soil cores, each with a diameter of 9 cm and a height of 4 cm, were obtained from the dye-stained and dye-unstained areas in each soil layer from the six experimental sites for measurement of the saturated hydraulic conductivity (SHC),

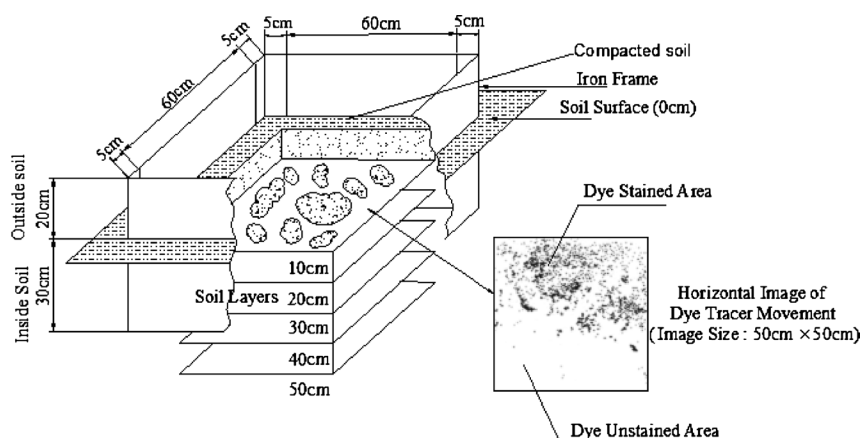


FIG. 3. Sketch map of the dye experiment.

which was determined using the constant-head method (Wang et al., 2010).

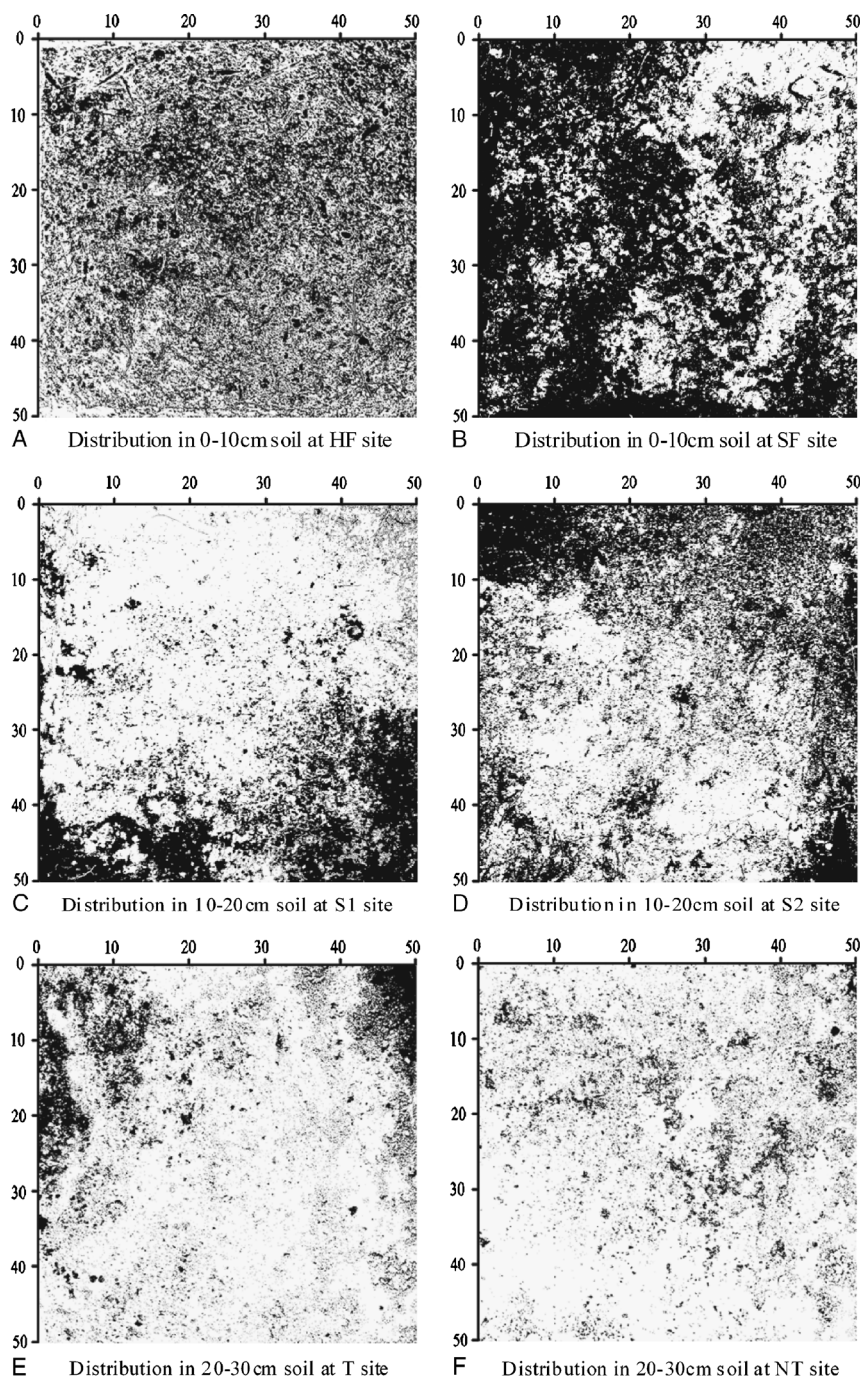
Root Hole and Root Length Measurements

Before the soil sampling, the number of root holes that could be observed by the naked eye (with a diameter >5 mm) was recorded for each soil layer. The roots were then sieved and washed with tap water in a 0.5-mm sieve. The root length

(in cm), surface area (in mm^2), diameter (in mm), and volume (in cm^3) were recorded. Each parameter was calculated for all of the root samples with diameters of less than 1, 1 to 3, 3 to 5, 5 to 10, and greater than 10 mm.

RESULTS AND DISCUSSIONS

The present study aimed to reveal the distribution of preferential flow paths and to evaluate the effect of the land use and



Note: HF means hardwood forest site. SF means softwood forest site.
S1 means the first shrub land site. S2 means the second shrub land site.
T means the tilled farmland site. NT means the non-tilled farmland site.

FIG. 4. Images of the dye-stained soils at different sites.

TABLE 1. Average Percentage of Horizontal Dye-Stained Area

Layers, cm	HF	SF	S1	S2	T	NT
0–10	54.47 (5.08)	63.09 (5.03)	79.63 (5.32)	81.87 (4.76)	75.92 (2.26)	63.99 (3.05)
10–20	45.48 (4.54)	40.87 (3.74)	48.13 (3.62)	54.32 (7.12)	54.34 (2.91)	33.17 (1.87)
20–30	26.37 (4.41)	27.47 (2.87)	29.39 (6.10)	26.65 (2.52)	19.92 (2.11)	11.45 (1.04)
30–40	17.13 (1.56)	18.31 (1.88)	14.28 (2.07)	15.99 (2.68)	9.60 (1.39)	6.74 (1.01)
40–50	8.12 (0.57)	7.18 (0.91)	7.09 (1.26)	4.95 (2.11)	3.41 (0.85)	3.66 (0.66)

Values in parentheses are standard deviations.

HF: hardwood forest; NT: no till; T: tilled.

parameters, such as the soil properties, root length per soil volume, root holes, and root surface area, on the occurrence of the preferential flow paths.

Horizontal Distribution of Preferential Flow Paths

The images of each horizontal soil layer were taken and corrected by Adobe Photoshop CS3, and the percentage of the stained area was calculated by Image-Pro Plus 6.0. Some of the images of the horizontal cross sections under the three land-use types are shown in Fig. 4. The percentages of the dye coverage are shown in Table 1.

The horizontal dye investigation revealed that all of the sites differed distinctly in both preferential flow path type and size. At the experimental sites, different types of macropores, such as worm burrows, root channels, and interaggregate voids (cracks and fissures), were observed. Both of the shrubland sites had a characteristic dominance of worm burrows and root channels. Both of the forestland sites showed very few worm burrows but did show a strong dominance of root channels that were both vertically and horizontally oriented. At both farmland sites, the dominant macropore types were interaggregate voids, which resulted from tillage and harvesting in the topsoil and root channels in the deeper layer.

The dye tracer experiments also revealed that the percentage of the preferential flow paths differed greatly among the studied sites. The variability within the sites of one land-use type was generally smaller than the variability among the sites.

The common point among the different sites of all of the land-use types was that the uppermost 0 to 10 cm of the soil had a dye coverage of greater than 50%, and the dye-stained area rapidly decreased below this depth. Below a depth of 20 cm, the dye-stained area was less than 30% for all of the land-use types (Table 1), confirming the results of Yasuda et al. (2001), who have reported that only the upper 0 to 10 cm of T soil had a nearly complete dye coverage and that, below a depth of 15 cm, the dye coverage was less than 50%. Gjettermann et al. (1997) have also found that, in the topmost 0 to 25 cm, the degree of dye coverage tended to be larger for lower irrigation intensities, indicating that the water flow in the topsoil took place via several pores in the soil matrix. In addition, Ohrstrom et al. (2002) have found that the dye coverage was almost 100% in the uppermost 5 to 10 cm and that it decreased rapidly with depth.

The distribution pattern of preferential flow paths is affected by various factors. Although the three land-use types had the same tendency in the distribution of the preferential flow paths, there were large differences among them. There were more preferential flow paths in the topsoil (0–20 cm) of the first shrubland site, the second shrubland site, and the T farmland site than in the topsoil of the forestland. At a soil depth of

20 to 30 cm, there was no difference between the forestland and the shrubland, and both had more preferential flow paths than the farmland. In the deeper soil depth (30–50 cm), however, there were more preferential flow paths in the forestland than in the shrubland and farmland. This condition is due to the occurrence of the preferential flow paths being closely related to the root characteristics (especially decayed root holes), soil cracks, and fractures (Lakshmikantha et al., 2009). The topsoil (0–20 cm) of the farmland may have numerous fissures and cracks, which can induce preferential flow paths. Most of the roots found in the shrubland and farmland were in shallow soil layers. In contrast, most of the forestland roots were in deep soil layers, which was why the HF site had more preferential flow paths than the shrubland and farmland in the deep soil layers. Lesturgez et al. (2004) found that grass can create macropores in the compact layer of the soil, which may also explain why the shrubland had more preferential flow paths in the topsoil (0–20 cm) than the forestland. The present result agrees with the conclusion of Hangen et al. (2002) that a high-stain percentage at a 5-cm soil depth reflects the macropore flow paths along horizontally oriented cracks; the distributing effect of the crop roots on the percolation patterns in the topsoil was not found in the study of Hangen et al. (2002).

Relationship Between Soil Properties and Preferential Flow Paths

The soil nutrient concentrations and the SHC in different land-use types are shown in Fig. 5. The soil nutrients varied little within and among the sites. However, the SHC varied markedly, demonstrating a large difference in the soils among the sites (Price et al., 2010).

The preferential flow paths have physicochemical properties that are distinct from those of the soil matrix. Comparing the dye-stained and dye-unstained areas, Fig. 5 shows that, regardless of the land-use type, the SHC was higher in the dye-stained area than in the dye-unstained area. The average SHC in the dye-stained areas was 41.3 mm min⁻¹ compared with 21.4 mm min⁻¹ in the dye-unstained areas. The SHC was significantly related to the dye-stained area (Table 2). The higher SHC in the dye-stained area was probably caused by the preferential flow paths, which can accelerate water movement, confirming the conclusion of Bundt et al. (2001) that the preferential flow paths actively conduct water.

The soil nutrients (available potassium, available phosphorus, and total nitrogen) showed no particular tendency between the dye-stained and dye-unstained areas. The soil organic matter content was approximately 5% to 20% higher in the dye-stained area than that in the dye-unstained area. This value was much lower than that obtained by Bundt et al. (2001), who reported 10% to 70% higher values in the preferential flow paths than in the matrix. The higher organic matter content in the preferential

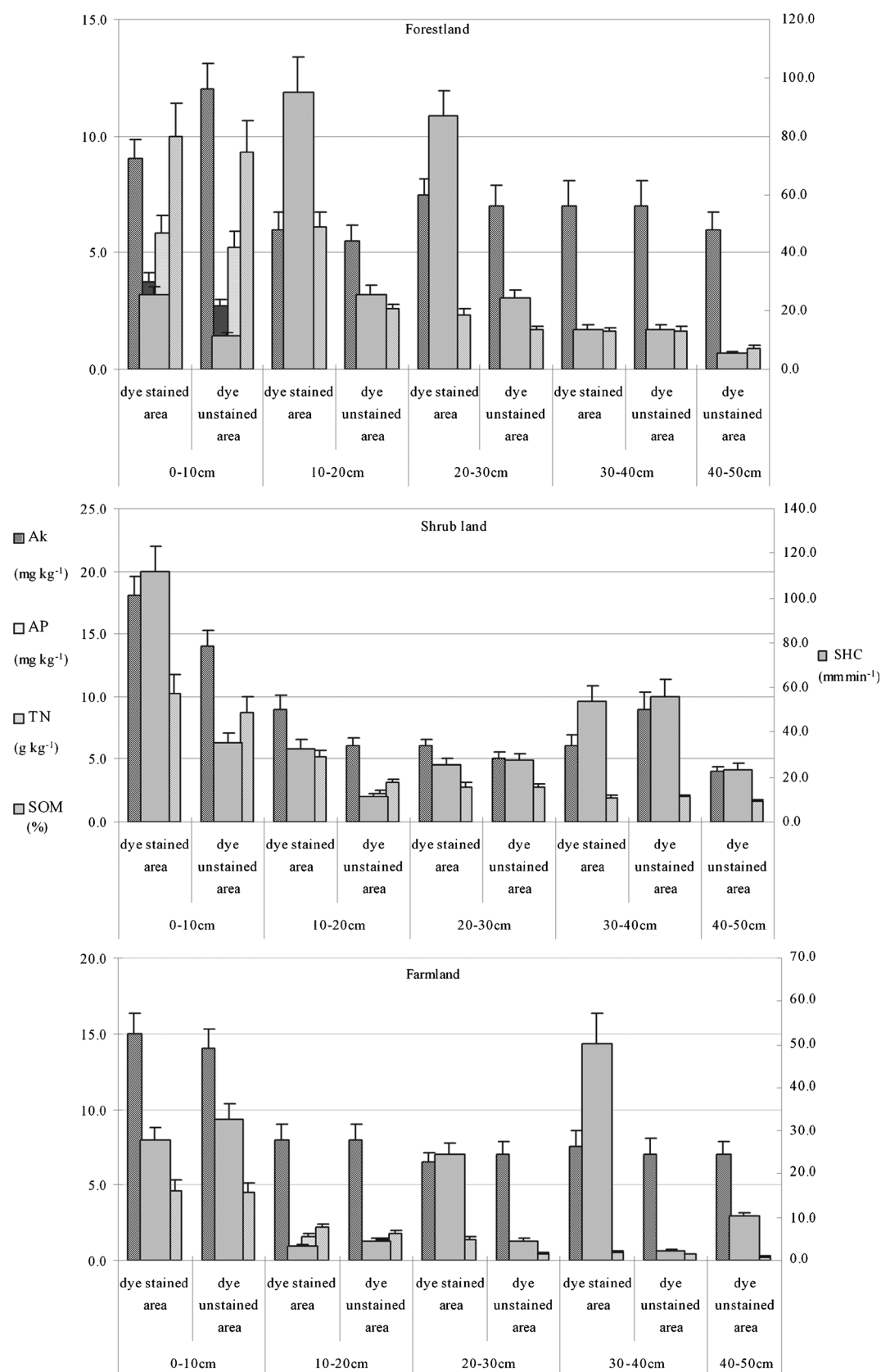


FIG. 5. Soil nutrient concentrations and the SHC values in different land-use types.

TABLE 2. Correlations Between Root Lengths, Root Holes, Root Surface Area, and Dye-Stained Area

	Stained Area	Length of Root <1 mm	Length of Root 1–3 mm	Length of Root 3–5 mm	Length of Root 5–10 mm	Length of Root >10 mm	Holes	Root Surface Area	SHC
Stained area	1	0.673*	0.658*	0.674 [†]	0.478 [†]	0.437	0.667*	0.617*	0.775*
Length of root <1 mm		1	0.941*	0.816*	0.805*	0.576 [†]	0.323	0.308	0.324
Length of root 1–3 mm			1	0.785*	0.724*	0.448 [†]	0.427	0.512 [†]	0.543 [†]
Length of root 3–5 mm				1	0.812*	0.453	0.817*	0.706*	0.721*
Length of root 5–10 mm					1	0.782*	0.967*	0.917*	0.895*
Length of root >10 mm						1	0.928*	0.912*	0.912*
Holes							1	0.914*	0.931*
Root surface area								1	0.804*
SHC									1

*Correlation is significant at the 0.01 level (2-tailed).

[†]Correlation is significant at the 0.05 level (2-tailed).

SHC: saturated hydraulic conductivity.

flow paths could be caused by the larger percentage of decayed roots in the preferential flow paths than in the soil matrix (Bundt et al., 2000).

Root Characteristics and Their Effects on Preferential Flow Paths

Root Characteristics

The root lengths per soil volume in the various land-use types are shown in Fig. 6, which provides an overview of the root length per soil volume. The variation in the root length per soil volume is considerable, ranging from 5,407 to 12.8 cm cm⁻³. The root length per soil volume in the different land-use types demonstrated the same tendency, and all of the land-use types had more roots with a diameter of less than 1 mm than the roots of other diameters. The descending order of root length per soil volume was as follows: less than 1-, 1- to 3-, 3- to 5-, 5- to 10-, and greater than 10-mm diameters. The root length per soil volume with diameters of less than 1, 1 to 3, and 3 to 5 mm reached a maximum in the topsoils of the SF, S1, S2, T, and NT plots (Fig. 6). The root length per soil volume decreased with the soil depth in all of the land-use types except for the HF plot.

Correlations of the Preferential Flow Path Percentage With the Root Length Per Soil Volume, Root Holes, and Root Surface Area

The relationship of the preferential flow path percentage to the root length density, root holes, and root surface area was tested using the partial correlation method in SPSS 16.0 (Table 2). SPSS is a computer program for statistical analysis, with its first version released in 1968. This partial correlation method is superior to other simple correlation methods because it can eliminate the effects of the factors on each other.

Table 2 shows that the dye-stained area is significantly related to the root length per soil volume with less than 1-, 1- to 3-, and 3- to 5-mm root diameters, root holes, and root surface area ($P < 0.01$). This condition supports the hypothesis of Lange et al. (2009) that tree roots are a key factor in the preferential infiltration in soils and the hypothesis of Germann et al. (2007) that tree roots are linked to preferential flow parameters. Noguchi et al. (1997) noted that at least 70% of the macropores (>2 mm) in the topsoil and 55% in the sub-soil of a forest soil in Japan are associated with roots. These observations indicate

the importance of tree roots in the formation of preferential flow paths.

Table 2 also shows that the dye-stained area is not significantly related to the root length per soil volume with 5- to 10- and greater than 10-mm root diameters. Materechera et al. (1994) showed that roots create pores and also can fragment the soil aggregates through penetration, especially for coarse roots. The cracks adjacent to living alfalfa roots have only a temporary effect on the soil macropores, whereas decaying roots produce stable macropores (Mitchell et al., 1995). The living coarse roots certainly cannot produce stable macropores. The turnover rates of fine roots (with a diameter of <2 mm) from spruce, fir, and beech were determined as approximately 0.7, 1.1, and 0.6 per year, respectively, and the mean turnover rate of the coarse roots (diameter >5 mm) was 3.4 per year (Withington et al. 2006). These results indicate that fine roots may produce more macropores with a diameter of less than 2 mm than coarse roots during the same period, also indicating that the fine roots have more significant effects on the preferential flow paths than the coarse roots.

Effects of Land Use on Preferential Flow Paths

The dye tracer experiments indicated that the distribution and characteristics of the preferential flow paths differed among land-use types. Given that all of the soils had similar textures, the variation in the preferential flow paths must be attributed to other factors, such as soil structure, soil physical and chemical properties, and so on. Land use has intense effects on the development of the soil structure (Motzki et al., 1996). Hangen et al. (2002) showed that some methods of conservation tillage, such as covering with mulch, would prevent the transport of a tracer beneath a 5-cm soil depth, whereas the water would reach a depth of 20 cm at conventionally T sites. Land management practices, such as the selection of plant species, plant densities, or tillage methods, may influence the soil properties and root distribution and thus indirectly influence the infiltration properties. These findings suggest that land use is important in the distribution and characteristics of preferential flow paths.

CONCLUSIONS

The distributions of the preferential flow paths in different land-use types had the same tendency in the distribution of the preferential flow paths. A greater than 50% dye coverage was

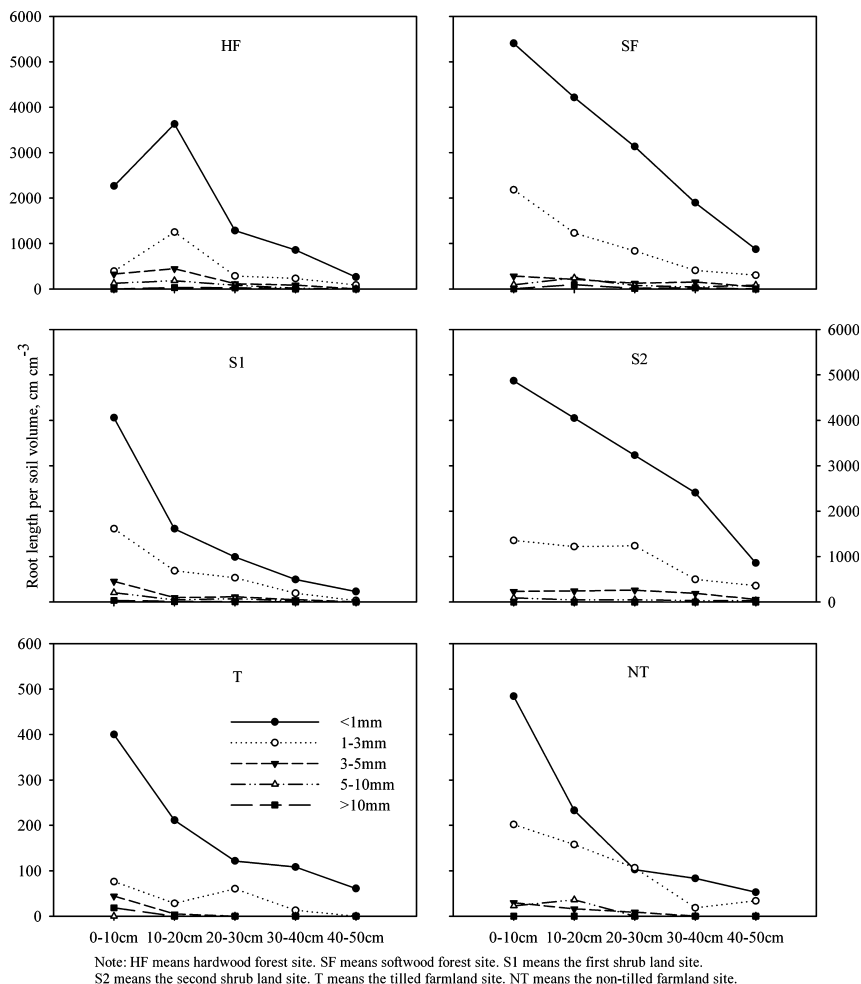


FIG. 6. Root length per soil volume at the different sites.

found in the uppermost 0 to 10 cm of the soil in all of the land-use types, but it decreased rapidly with depth.

The soil chemical properties and the SHC varied between the preferential flow path and soil matrix. The soil area with preferential flow paths had a larger SHC than that without preferential flow paths, with the average value of 41.3 and 21.4 mm min⁻¹, respectively. The soil organic matter content was approximately 5% to 20% higher in the preferential flow paths than in the soil matrix.

Soil properties and root characteristics are important in the formation of preferential flow paths. The present study reveals the relationship of the soil properties, root characteristics, and preferential flow paths. The percentage of preferential flow paths in the soil was significantly affected by the root length per soil volume with less than 1-, 1- to 3-, and 3- to 5-mm root diameters, but it was not significantly affected by the root density with 5- to 10-mm and greater than 10-mm root diameters, indicating that the roots with a diameter of less than 5 mm affect the formation of the preferential flow paths.

Different land-use types have led to a wide variety of soil properties and root distributions, thus to different preferential flow path distributions. In the present study, the shrubland and the farmland had more preferential flow paths in the topsoil than the forestland. Conversely, there were more preferential flow paths in the forestland than in the shrubland and farmland

in the deeper soil layers. The current study has contributed to a better understanding of the significant effects of land-use types on the preferential flow paths in soils.

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